

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Name

Dynamic Multi-Tenant Form Management & Reporting Platform

Version

v1.0

Project Type

Enterprise Web Application

Purpose

To provide a platform where Super Admins can create dynamic forms and distribute them across Centers, Organizations, and Users while supporting role-based access control, conditional question flows, dynamic reporting, OTP verification, and organization-specific customization.

Problem Statement

Organizations currently use static forms that are difficult to manage, update, and distribute.

The platform should:

- Allow centralized form management.
- Allow controlled distribution of forms.
- Support dynamic form creation.
- Support conditional questions and workflows.
- Support automated reporting.
- Eliminate hardcoded form structures.

Business Goals

Goal 1

Create forms once and distribute everywhere.

Goal 2

Allow organizations to customize forms without affecting the original form.

Goal 3

Automate report generation.

Goal 4

Support thousands of organizations and users.

Goal 5

Provide a fully configurable and modular system.

User Roles

Super Admin

Highest authority.

Permissions

- Create forms
- Edit forms
- Delete forms
- Publish forms
- Make forms public/private
- Manage centers
- View all reports
- Manage permissions

Center

Controls organizations.

Permissions

- Create organizations
- Disable organizations
- Restrict organization access
- View organization reports

Organization

Controls users.

Permissions

- Add users
- Remove users
- Disable users
- Customize assigned forms
- View organization reports

User

End user.

Permissions

- Register
- Login
- Verify OTP
- Fill forms
- Submit forms
- View reports

Functional Requirements

Authentication Module

Login Methods

- Email Login
- Google Login
- Phone Login
- Guest Login

Registration

User can register under an organization.

OTP Verification

User must verify OTP before accessing reports.

User Management Module

Super Admin

Can:

- Create centers
- Delete centers
- Update centers

Center

Can:

- Create organizations
- Disable organizations

Organization

Can:

- Create users

- Disable users

Form Management Module

Dynamic Form Builder

System shall allow:

- Create forms
- Create sections
- Create questions
- Create options
- Create validation rules

Supported Field Types

- Text
- Text Area
- Number
- Date
- Email
- Phone
- Radio
- Checkbox
- Dropdown
- File Upload
- Signature

Form Visibility

Public Form

Visible to:

- Centers
- Organizations
- Users

Private Form

Visible only to Super Admin.

Organization Form Customization

Organizations must not modify the original form.

Workflow:

Master Form

↓

Clone Form

↓

Organization Form

Organization edits only the cloned version.

Conditional Logic Engine

System shall support question dependency.

Example:

Question:

Do you own a vehicle?

Options:

- A. Car
- B. Bike
- C. None

If Car selected:

Show:

- Car Model
- Registration Number

If Bike selected:

Show:

- Bike Model
- Engine CC

If None selected:

Hide all dependent questions.

Dynamic Rule Engine

Rules supported:

- Show
- Hide
- Enable
- Disable
- Required
- Optional

Example:

IF Age > 18
THEN Show Driving License Question

Submission Module

Users can:

- Save Draft
- Submit Form

System must:

- Validate answers
- Store responses
- Trigger report generation

Reporting Module

Supported Formats:

- HTML
- PDF
- Excel

Report Filters

Support:

- Date Range
- User
- Organization
- Center
- Form
- Status

Notifications Module

Support:

- OTP Emails
- Verification Emails
- System Notifications

Role Based Access Control (RBAC)

Super Admin

Full access.

Center

Center-level access only.

Organization

Organization-level access only.

User

Own records only.

Non Functional Requirements

Performance

Page Load Time:

< 3 seconds

Scalability

Support:

- 10,000+ Users
- 1,000+ Organizations
- 100+ Centers

Security

- JWT Authentication
- Password Hashing
- OTP Verification
- RBAC

Availability

Target:

99.9% uptime

Maintainability

Requirements:

- Modular Architecture
- No Hardcoding

- Reusable Components

Technical Requirements

Frontend

- Vite
- Vanilla JavaScript
- HTML
- CSS/Tailwind

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

System Architecture

Super Admin



Center



Organization



User



Dynamic Form



Submission



Reports

Database Collections

Users

Stores all users.

Roles

Stores system roles.

Permissions

Stores system permissions.

Centers

Stores centers.

Organizations

Stores organizations.

Forms

Stores form metadata.

Questions

Stores questions.

QuestionOptions

Stores options.

ConditionalRules

Stores conditional logic.

FormAssignments

Stores form distribution.

Submissions

Stores form submissions.

SubmissionAnswers

Stores user answers.

Reports

Stores generated reports.

OTP

Stores OTP records.

Success Metrics

Adoption

- Number of organizations onboarded
- Number of users registered

Engagement

- Forms submitted
- Reports generated

System Health

- API response time
- Error rate
- Login success rate

Future Enhancements (Phase 2)

- Advanced Conditional Workflows
- WhatsApp OTP
- Audit Logs
- Approval Workflow
- AI-Based Analytics
- Dashboard Insights

- Multi-Language Support
- Digital Signatures

Future Enhancements (Phase 3)

- Workflow Engine
- BPM Integration
- Real-Time Notifications
- Multi-Tenant SaaS Deployment
- Mobile Application
- AI Form Recommendations

Acceptance Criteria

Project is considered complete when:

- Dynamic forms can be created.
- Forms can be published.
- Organizations can customize forms.
- Users can submit forms.
- Conditional questions work.
- Reports generate successfully.
- OTP verification works.
- RBAC works correctly.
- Public/Private form visibility works.
- System is fully API-driven.

Server Architecture:

src/

|

|—— app.ts

```
|—— server.ts
|
|—— config/
|   |—— db.ts
|   |—— env.ts
|   |—— jwt.ts
|   |—— mail.ts
|   |—— logger.ts
|
|—— modules/
|   |
|   |—— auth/
|       |—— auth.controller.ts
|       |—— auth.service.ts
|       |—— auth.repository.ts
|       |—— auth.routes.ts
|       |—— auth.validation.ts
|       |—— auth.types.ts
|       |
|       |—— users/
|           |—— user.controller.ts
|           |—— user.service.ts
|           |—— user.repository.ts
|           |—— user.routes.ts
|           |—— user.validation.ts
|           |—— user.types.ts
|           |
|           |—— centers/
```

```
| | └── center.controller.ts
| | └── center.service.ts
| | └── center.repository.ts
| | └── center.routes.ts
| | └── center.validation.ts
| |
| └── organizations/
| | └── organization.controller.ts
| | └── organization.service.ts
| | └── organization.repository.ts
| | └── organization.routes.ts
| | └── organization.validation.ts
| |
| └── forms/
| | └── form.controller.ts
| | └── form.service.ts
| | └── form.repository.ts
| | └── form.routes.ts
| | └── form.validation.ts
| | └── form.types.ts
| |
| └── questions/
| | └── question.controller.ts
| | └── question.service.ts
| | └── question.repository.ts
| | └── question.routes.ts
| |
| └── submissions/
```

```
| | └── submission.controller.ts
| | └── submission.service.ts
| | └── submission.repository.ts
| | └── submission.routes.ts
| |
| └── reports/
| | └── report.controller.ts
| | └── report.service.ts
| | └── report.repository.ts
| | └── report.routes.ts
| | └── pdf.generator.ts
| | └── excel.generator.ts
| |
| └── otp/
| | └── otp.controller.ts
| | └── otp.service.ts
| | └── otp.repository.ts
| | └── otp.routes.ts
| |
| └── notifications/
| | └── notification.service.ts
| | └── email.service.ts
| | └── sms.service.ts
| |
| └── permissions/
| | └── permission.service.ts
| | └── role.service.ts
| | └── permission.routes.ts
```



```
|
|----- database/
|
| |
| |----- models/
| |
| | |----- User.model.ts
| |
| | |----- Center.model.ts
| |
| | |----- Organization.model.ts
| |
| | |----- Form.model.ts
| |
| | |----- Question.model.ts
| |
| | |----- Submission.model.ts
| |
| | |----- Report.model.ts
| |
| | |----- OTP.model.ts
| |
| | |----- Role.model.ts
| |
| | |----- Permission.model.ts
| |
| |
| |----- index.ts
|
|----- middleware/
|
| |----- auth.middleware.ts
|
| |----- role.middleware.ts
|
| |----- validation.middleware.ts
|
| |----- error.middleware.ts
|
| |----- upload.middleware.ts
|
|----- shared/
|
| |
| |----- constants/
|
| |----- roles.ts
```

```
| | | |── permissions.ts
| | | |── status.ts
| | |
| | |── helpers/
| | | |── generateOtp.ts
| | | |── generateToken.ts
| | | |── hashPassword.ts
| | | |── comparePassword.ts
| | |
| | |── utils/
| | | |── apiResponse.ts
| | | |── pagination.ts
| | | |── date.ts
| | | |── logger.ts
| | |
| | |── types/
| | | |── api.types.ts
| | | |── express.types.ts
| |
| |── routes/
| | |── index.ts
```

Client Architecture:

```
client/
|
|── public/
| |── favicon.ico
| |── robots.txt
```

```
|
|
|—— src/
|
|
|   |—— main.js
|   |—— app.js
|
|   |—— assets/
|   |
|   |   |—— images/
|   |   |
|   |   |   |—— logo.png
|   |   |   |—— login-banner.png
|   |   |   |—— avatars/
|   |   |
|   |   |—— icons/
|   |   |
|   |   |   |—— dashboard.svg
|   |   |   |—— forms.svg
|   |   |   |—— reports.svg
|   |   |
|   |   |—— fonts/
|   |
|
|   |—— router/
|   |
|   |   |—— router.js
|   |   |—— routes.js
|   |   |—— protectedRoute.js
|   |   |—— roleRoute.js
|   |
|
|   |—— layouts/
|   |
|   |   |—— AuthLayout.js
|   |   |—— DashboardLayout.js
```

```
| | | └── SuperAdminLayout.js
| | | └── CenterLayout.js
| | | └── OrganizationLayout.js
| | | └── UserLayout.js
| |
|
| └── views/
| |
| | └── auth/
| | |
| | | └── LoginView.js
| | | └── RegisterView.js
| | | └── VerifyOtpView.js
| | | └── ForgotPasswordView.js
| | | └── ResetPasswordView.js
| | |
| |
| | └── super-admin/
| | |
| | | └── DashboardView.js
| | | └── CentersView.js
| | | └── OrganizationsView.js
| | | └── UsersView.js
| | | └── FormsView.js
| | | └── ReportsView.js
| | | └── SettingsView.js
| | |
| |
| | └── center/
| | |
| | | └── DashboardView.js
| | | └── OrganizationsView.js
```

```
| | | |— UsersView.js
| | | |— ReportsView.js
| | | |— SettingsView.js
| | |
| |
| | |— organization/
| | |
| | | |— DashboardView.js
| | | |— UsersView.js
| | | |— FormsView.js
| | | |— ReportsView.js
| | | |— SettingsView.js
| | |
| |
| | |— user/
| |
| | |— DashboardView.js
| | |— MyFormsView.js
| | |— MyReportsView.js
| | |— ProfileView.js
| | |— SettingsView.js
| |
|
| |— modules/
| |
| | |— auth/
| | |
| | | |— auth.service.js
| | | |— auth.store.js
| | | |— auth.validation.js
| | | |— auth.events.js
| | | |— auth.constants.js
| | |
```

```
| |
| | └── users/
| |
| | |
| | | └── user.service.js
| | | └── user.validation.js
| | | └── user.events.js
| | | └── user.constants.js
| | |
| |
| |
| | └── centers/
| |
| | |
| | | └── center.service.js
| | | └── center.validation.js
| | | └── center.events.js
| | |
| |
| |
| | └── organizations/
| |
| | |
| | | └── organization.service.js
| | | └── organization.validation.js
| | | └── organization.events.js
| | |
| |
| |
| | └── forms/
| |
| | |
| | | └── form.service.js
| | | └── form.validation.js
| | | └── form.events.js
| | | └── form.constants.js
| | | └── form.state.js
| | |
| |
| |
| | └── reports/
```

```
| | |
| | | |—— report.service.js
| | | |—— pdf.service.js
| | | |—— excel.service.js
| | | |—— report.filters.js
| | |
| |
| | |—— submissions/
| | |
| | | |—— submission.service.js
| | | |—— submission.validation.js
| | | |—— submission.state.js
| | |
| |
| | |—— otp/
| | |
| | | |—— otp.service.js
| | | |—— otp.events.js
| | | |—— otp.validation.js
| | |
| |
| | |—— permissions/
| |
| | |—— role.service.js
| | |—— permission.service.js
| | |—— permission.guard.js
| |
|
| |—— dynamic-form/
| |
| | |—— builder/
| | |
| | | |—— FormBuilder.js
| | | |—— SectionBuilder.js
```

```
| | | |─ QuestionBuilder.js
| | | |─ OptionBuilder.js
| | | |─ RuleBuilder.js
| | |
| |
| | |─ renderer/
| | |
| | | |─ FormRenderer.js
| | | |─ SectionRenderer.js
| | | |─ QuestionRenderer.js
| | | |─ OptionRenderer.js
| | | |─ ConditionalRenderer.js
| | |
| |
| | |─ validators/
| | |
| | | |─ requiredValidator.js
| | | |─ regexValidator.js
| | | |─ minValidator.js
| | | |─ maxValidator.js
| | |
| |
| | |─ state/
| | |
| | | |─ formStore.js
| | | |─ questionStore.js
| | | |─ answerStore.js
| | |
| |
| | |─ rules/
| | |
| | | |─ conditionalEngine.js
| | | |─ visibilityEngine.js
```



```
| | | └── validationEngine.js
| | | └── dependencyEngine.js
| |
|
| | └── components/
| |
| | | └── navbar/
| | | | └── Navbar.js
| | | | └── navbar.css
| | |
| | | └── sidebar/
| | | | └── Sidebar.js
| | | | └── sidebar.css
| | |
| | | └── modal/
| | | | └── Modal.js
| | | | └── modal.css
| | |
| | | └── table/
| | | | └── Table.js
| | | | └── table.css
| | |
| | | └── loader/
| | | | └── Loader.js
| | | | └── loader.css
| | |
| | | └── toast/
| | | | └── Toast.js
| | | | └── toast.css
| |
|
| | └── services/
| |
| | | └── api.js
```

```
| | |—— http.js
| | |—— storage.js
| | |—— token.service.js
| |
|
| |—— constants/
| |
| | |—— roles.js
| | |—— permissions.js
| | |—— routes.js
| | |—— statuses.js
| | |—— apiEndpoints.js
| |
|
| |—— utils/
| |
| | |—— date.js
| | |—— formatter.js
| | |—— download.js
| | |—— debounce.js
| | |—— throttle.js
| | |—— helpers.js
| |
|
| |—— styles/
| |
| | |—— globals.css
| | |—— variables.css
| | |—— layout.css
| | |—— dashboard.css
| | |—— forms.css
| | |—— reports.css
|
```

|
|—— index.html
|—— vite.config.js
└—— package.json

Database Structure

Users
Roles
Permissions
RolePermissions
Centers
Organizations
Forms
Questions
QuestionOptions
ConditionalRules
FormAssignments
Submissions
SubmissionAnswers
Reports
OTPs

Field Types:-

TEXT
TEXTAREA
NUMBER
EMAIL
PHONE
DATE
TIME
RADIO
CHECKBOX
DROPDOWN
FILE
SIGNATURE